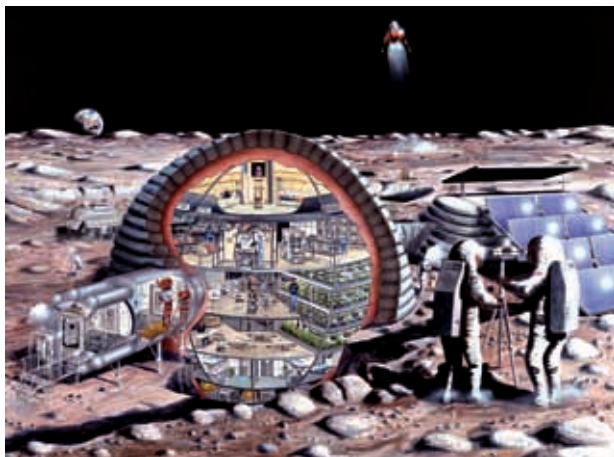


While it isn't actually a Space Race, the two targets that everyone seems to want to achieve is the colonisation of moon and planning man's first visit another planet in the solar system – Mars.

Lunar colonies

Ever since the Apollo 11, every space agency has harboured plans of going to the moon and setting up a colony there. After the Moon Mineralogy Mapper aboard the Chandrayaan-I discovered water ice on the moon, this has gotten a bit more of a push from all parties.

Of course, no one is close to actually achieving it, even though Russia and Japan have announced plans to do so by 2025 and China has said that it plans



An artist's concept of a lunar outpost in the future

to do it by 2030. NASA was looking at lunar colonisation as part of the Constellation programme, but that was cancelled in 2010.

So far, early plans of every programme seem to be looking at inflation-deployed expandable structures as the most likely building block for a lunar base. Inflatables can be used as connectors or tunnels between crew quarters and can provide radiation shelter if covered with lunar soil/

Before the cancellation of Constellation, NASA had previously worked on a 12-foot-diameter inflatable structure made of multilayer fabric, housing technologies such as flexible structural health monitoring systems, self-healing materials and radiation protective materials. Attached to the structure is a smaller inflatable structure that serves as a demonstration airlock. Both are essentially pressurized cylinders, connected by an airtight door.